India Runs – Data & AI Challenge

Redrob AI Ranker: Intelligent Candidate Discovery & Ranking

Team TOPGUN

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How to Run the Project

Prerequisites: Python 3.8+ and any modern browser

Step 1 – Run the Ranking Engine (CLI):

```
python run.py --candidates candidates.jsonl --out team_submission.csv
```

This reads the full dataset, scores all 100,000+ candidates in parallel, and writes the final `team_submission.csv`. **Step 2 – Launch the Dashboard (UI):**

```
python -m http.server 8000
Then open: http://localhost:8000/frontend/index.html
```

Upload `team_submission.csv` when prompted to view the interactive recruiter dashboard.

Executive Summary

- **Objective:** Build an intelligent candidate discovery and ranking system for the Redrob platform.
- **Core Philosophy:** Rank candidates based on semantic meaning, career trajectory, and behavioral intent – not keyword frequency.
- **Performance:** Process 100,000+ candidates in under 35 seconds using multiprocessing.
- **Architecture:** Fully offline, deterministic, and production-grade – zero API calls, zero external dependencies.
- **Transparency:** Every ranked candidate comes with an Explainable AI (XAI) rationale.

The Problem Statement

The Limitations of Traditional ATS

- **Keyword Reliance:** Traditional systems count how many times a term appears.
- **Lack of Context:** "Used Python in college" vs. "Architected a Python microservice" score identically.
- **Recruiter Fatigue:** Teams waste hours manually sifting through mis-ranked profiles.
- **Missed Talent:** Exceptional engineers who write concise resumes are buried beneath keyword-stuffed profiles.
- **Ignored Signals:** Platform signals like assessment scores, notice periods, and recruiter saves go completely unused.

Our Solution Overview

- **Semantic Understanding:** Skills grouped by meaning, not exact string matches.
- **Career Shape Analysis:** Trajectory of titles analyzed using the Needleman-Wunsch algorithm.
- **Full Signal Utilization:** Every field in the schema is used – education tier, assessment scores, notice period, recruiter saves, offer acceptance rate.
- **Intent Scoring:** Measures how actively and reliably the candidate is looking.
- **Anti-Gaming Integrity:** Statistically detects and penalizes artificial profile inflation.

The 4 Pillars of Intelligent Ranking

Our engine evaluates candidates across four weighted axes:

1. **Semantic Competency (40%):** Technical fit across clustered skill concepts.
2. **Career Trajectory (30%):** Pace of promotion, company pedigree, and education tier.
3. **Behavioral Intent (20%):** Engagement signals and readiness to join.
4. **Location Intelligence (10%):** Geographical alignment with the role.

$$\texttt{Final Score = (Skills} \times \texttt{0.40) + (Experience} \times \texttt{0.30) + (Behavior} \times \texttt{0.20) + (Location} \times \texttt{0.10)}$$

Pillar 1: Semantic Competency (40%)

Moving Beyond Exact String Matches

- **IDF-Weighted Cluster Matching:** Rare skills carry more weight than common ones.
- **Endorsement Bonus:** Skills with high peer endorsements receive up to a 15% IDF weight increase.
- **Assessment Score Integration:** Platform skill test scores (0–100) boost or penalize cluster match strength.
- **Full-Text Search:** Candidate summary and headline text are also searched – not just the skills list.
- **Cohesion Check:** Advanced skills without foundational programming knowledge incur a 20% discount.

Skill Clusters in Practice

Cluster	Accepted Technologies
Core Programming	Python, Go, C++, Java
Vector Databases	Pinecone, Weaviate, Qdrant, Milvus, FAISS
Retrieval & Ranking	BM25, Cross-encoders, NDCG, Hybrid Search
LLM Engineering	Fine-tuning, LoRA, LangChain, HuggingFace
Infrastructure	Docker, Kubernetes, FastAPI, AWS, GCP

Pillar 2: Career Trajectory (30%)

Promotion Velocity, Pedigree, and Education

- **Experience Banding:** Evaluates whether total years fit the Ideal, Acceptable, or Outside band.
- **Promotion Velocity:** Rewards rapid progression to Senior/Lead/Principal titles.
- **Education Tier:** Tier-1 institutions in relevant fields (CS, AI, Statistics) earn up to +12 points.
- **Company Size Signal:** Large-scale engineering experience (10,000+ employees) earns a moderate bonus.
- **Consulting Penalty:** Profiles from pure consulting firms are down-weighted when the role requires product-building instincts.

Dynamic Career Sequence Alignment

Applying Bioinformatics to Career Paths

- **Algorithm:** Adapts the Needleman-Wunsch sequence alignment algorithm from genomics.
- **Ideal Sequence:** [Junior $\rightarrow$ Mid $\rightarrow$ Senior $\rightarrow$ Lead $\rightarrow$ Principal]
- **Execution:** Aligns the candidate's chronological job titles against this ideal progression.
- **Scoring:**
  - Match Reward: $+1.0$
  - Gap Penalty: -0.5 (stagnation)
  - Mismatch Penalty: proportional to level distance

Corporate Pedigree via PageRank

- **The Concept:** Not all work experience is equal. Selective companies produce stronger signals.
- **Implementation:** A directed transition graph $(Company_A \rightarrow Company_B)$ is built across the entire 100,000-candidate pool.
- **PageRank Execution:** Authority of each organization is determined dynamically – no hardcoded lists.
- **Result:** Candidates from high-authority nodes receive a strategic, data-driven score boost.

Pillar 3: Behavioral Intent (20%)

Measuring Genuine Engagement – All Signals Used

- **Recruiter Response Rate:** Historically responsive candidates rank higher.
- **Notice Period:** $\leq$15 days earns a strong boost; $>$90 days incurs a penalty.
- **Interview Completion Rate:** Candidates who ghost interviews are penalized.
- **Offer Acceptance Rate:** Candidates with a history of rejecting offers are flagged.
- **Saved by Recruiters (30d):** 5+ saves signals strong external market validation.
- **Average Response Time:** Fast responders (under 4 hours) are boosted.
- **Platform Activity Decay:** Exponential time-decay penalizes inactive profiles.

Pillar 4: Location Intelligence (10%)

Geographical Alignment

- **Preferred Cities (e.g., Pune, Delhi NCR):** Full location score (1.0).
- **Acceptable Cities (e.g., Bangalore, Mumbai):** Moderate score (0.7).
- **Willing to Relocate:** Candidates outside preferred zones but open to relocation receive a baseline score (0.6).
- **Remote / Unlisted:** Minimum baseline (0.2) – not eliminated, but ranked lower.

- **Impossible Skill Claims:** "Expert" proficiency + 0 months usage = instant disqualification.
- **Assessment Contradiction:** Claims "Expert Python" but scored 15/100 on the platform Python test = flagged.
- **Timeline Inconsistencies:** 10+ years stated but career history shows under 1 year = flagged.
- Flagged profiles are assigned a score of 0 and excluded from the final ranking.

Shannon Entropy Anomaly Detection

Statistical Profiling of Unnatural Resumes

- **The Theory:** Natural human language has a predictable vocabulary distribution.
- **The Math:** Shannon Entropy is calculated over the resume text.

$$H(X) = -\sum P(x_i) \log_2 P(x_i)$$

- **The Trigger:** Profiles deviating more than 2σ from the population mean are flagged automatically.
- **The Result:** Successfully catches machine-generated, keyword-stuffed, and white-fonted resumes.

High-Level System Architecture

- **Layer 1 – Parallelized Python Engine (Offline CLI):**
  - O(N) streaming reader – memory footprint under 200 MB.
  - Pre-Scan Phase: computes global TF-IDF weights, entropy baseline, PageRank graph.
  - Multiprocessing pool shards candidates across all CPU cores.
- **Layer 2 – Interactive Recruiter Sandbox (Frontend UI):**
  - Pure HTML/JS single-file application.
  - No Node.js, no build steps, no server required.

The Parallelized Scoring Engine

- **Speed:** Processes batches of candidates concurrently across all available CPU cores.
- **Benchmark:** 100,000+ candidate profiles ranked in **under 35 seconds** on a standard CPU.
- **Efficiency:** Executes within the 5-minute hackathon constraint ($8\times$ faster).
- **Deterministic Sort:** Ranked by $(-round(score, 4),\ candidate\_id)$ – identical results on every run.
- **Scale:** The same architecture handles millions of candidates with linear scaling.

Explainable AI (XAI)

Generating Deterministic Rationale

- **No Black Boxes:** Every shortlist placement is fully justified.
- **Context-Aware:** Rationale is built from actual matched parameters, not templates.
- **Example Output:**

  "Exceptional technical fit, covering 5/5 core semantic clusters. 6.4 years of steady progression. Tier-1 academic background. High engagement intent: open to work, saved by 26 recruiters in the last 30 days."

- **Trust:** Builds recruiter confidence in algorithmic decisions.

Fully Offline Operation

Meeting Strict Privacy and Latency Constraints

- **Zero API Calls:** Entire pipeline runs on Python's standard library.
- **No Cloud Dependency:** No OpenAI, no Gemini, no HuggingFace inference endpoints.
- **Security:** Candidate PII never leaves the local machine.
- **Cost:** Zero per-query cost – scales to any volume without incurring inference fees.
- Complies fully with sandbox environment restrictions.

Configurable for Any Role

- **Adaptability:** No code changes required to hire for a different role.
- **JSON-Driven:** All role requirements live in `data/jd_rules.json`.
- **Customizable Parameters:**
  - Target role title
  - Experience minimums and maximums
  - Semantic skill clusters and their constituent technologies
  - Preferred and acceptable locations
  - Tier-1 and consulting company lists

The Recruiter Dashboard UI

Zero Setup, Maximum Insight

- **Technology:** A single pure HTML/JS file. No Node.js, no build steps, no npm.
- **Workflow:** Role Setup → CSV Upload → Processing Animation → Results.
- **Candidate Cards:** Visual display of rank, composite score, and the AI rationale.
- **Click-to-Expand:** Detailed scoring breakdown with individual progress bars for Skills, Experience, Behavior, and Location.
- **Real-Time Search:** Filter candidates by ID or reasoning text instantly.
- **Export:** One-click CSV re-export of filtered results.

Active Learning Loop (Recruiter Feedback)

Continuous Improvement

- **Recruiter Feedback:** Recruiters can "Invite" or "Decline" any candidate from the dashboard.
- **Weight Adjustment:** A single-layer perceptron training step updates axis weights based on the recruiter's implicit preferences.
- **Adaptation:** Over time, weights shift to match a specific hiring manager's intuition – without manual tuning.
- **Vision:** Persistent preference profiles per recruiter for fully personalized ranking.

Submission Output

Official Hackathon Format

- The engine writes `team_submission.csv` following the official schema exactly.
- **Columns:** `candidate_id`, `rank`, `score`, `reasoning`
- **Rows:** Exactly 100 candidates, ranked 1 through 100.
- **Scores:** Non-increasing – no score inversions across ranks.
- **Validation:** Passed the organizers' official `validate_submission.py` with zero errors.

Performance Metrics & Constraints

Metric	Value
Dataset size	100,000+ candidate profiles
Processing time	$\sim$34 seconds
Memory usage	$<$ 200 MB RAM
External API calls	Zero
External dependencies	Zero (Pure Python stdlib)
Hackathon time limit	5 minutes (we are 8$\times$ faster)
Validation status	Passed – zero errors

Conclusion & Impact

- **For Recruiters:** Reduces time-to-shortlist from days to 35 seconds.
- **For Candidates:** Rewards genuine career growth, platform engagement, and real skills – not resume hacking.
- **For Redrob:** A production-ready, fully offline engine – adaptable to any JD with a single JSON configuration change.
- **Engineering Excellence:** Every available data signal is utilized, from education tier to interview completion rate.

Thank you.